

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		chlorine is more reactive than iodine (1) accept chlorine can gain an electron more easily than iodine award (1) for any of following - chlorine displaces iodide ion - chlorine takes electron from iodide ion - chlorine oxidises iodide ion	2			2		2
		(ii)		$\text{Cl}_2 + 2\text{KI} \rightarrow 2\text{KCl} + \text{I}_2$ award (1) for correct formulae for reactants and products award (1) for balancing only if correct formulae given		2		2		
	(b)			112 g of iron reacts with 213 g of chlorine (1) 1.32 g of iron reacts with $\frac{213}{112} \times 1.32$ g of chlorine (1) 2.51 g (1) ecf possible for incorrect relative mass values i.e. 112 or 213 alternative method moles Fe = $\frac{1.32}{56} = 0.0236$ mol (1) moles Cl ₂ = $\frac{3}{2} \times 0.0236 = 0.0353$ mol (1) mass Cl ₂ = $0.0353 \times 71 = 2.51$ g (1) ecf possible for incorrect mole ratio		3		3	3	

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		$\text{Cl}_2 + 3\text{Br}_2 \rightarrow 2\text{ClBr}_3$		1		1		
		(ii)		77.46 / 77.5 / 77		1		1	1	
				Question 7 total	2	7	0	9	4	2